

Visualization Assessment of Tendency in Clustering - $58K \times 58K$ in 60 seconds

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Abstract. Visualization Assessment of Tendency (VAT) in Clustering has been a technique for the identification of data clusters since 2002. Although a proven technique, it has suffered from the curse of dimensionality, leading to limited performance for datasets with more than 5000 elements. The core of the model is based on Prim's algorithm, a greedy Minimum Spanning Tree search that uses Bubble Sort at its core. By replacing Bubble Sort with Merge Sort using a priority queue, the updated "mergeVAT" algorithm complexity changes from $O(N) = N^3$ to $O(N) = N^2 \log N$. For datasets with 1,000 elements, this leads to a speed up of $30\times$. In addition, by identifying the permutation loops and bit-masking, it is now possible to compute the full VAT of a 58000 element dataset in place while requiring only $O(N) = \frac{N^2}{16}$ additional working memory in less than 2 minutes.

Keywords: VAT · Clustering · mergeVAT · Ant Colony Optimization

1 Background

1.1 VAT

VAT is a method for the visual identification of clusters in arbitrary dimension data sets. It converts the arbitrary dimensions into a 2D dissimilarity matrix by using a norm function. Most commonly, this is the L2-norm, also known as the euclidean norm. However, other norms may be used depending on the application and the desired measure of dissimilarity. This converts the N by K matrix of samples into a n by n matrix of pairwise distances, commonly called the dissimilarity matrix D. The core of the method is based upon a minimum spanning tree of the columns of the dissimilarity matrix D. More specifically, this method is very similar to Prim's algorithm. A greedy algorithm for computing the minimum spanning tree.

Since the original development of the VAT method[4], numerous alternate and custom implementations have been developed. Broadly speaking, these have attempted to address the issue of performance in Prim's algorithm. By enforcing other constraints on order, distance metric, or output, the methods improve